

Tennessee Tech PCB Design Workshop

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Workshop Overview

- PCB Design Principles and Practices (1 October) ← today!
- PCB Design Time in KiCad (6 October)
- PCB Design Time in KiCad (8 October)
- PCB Assembly and Testing (13 October*)

*tentative date

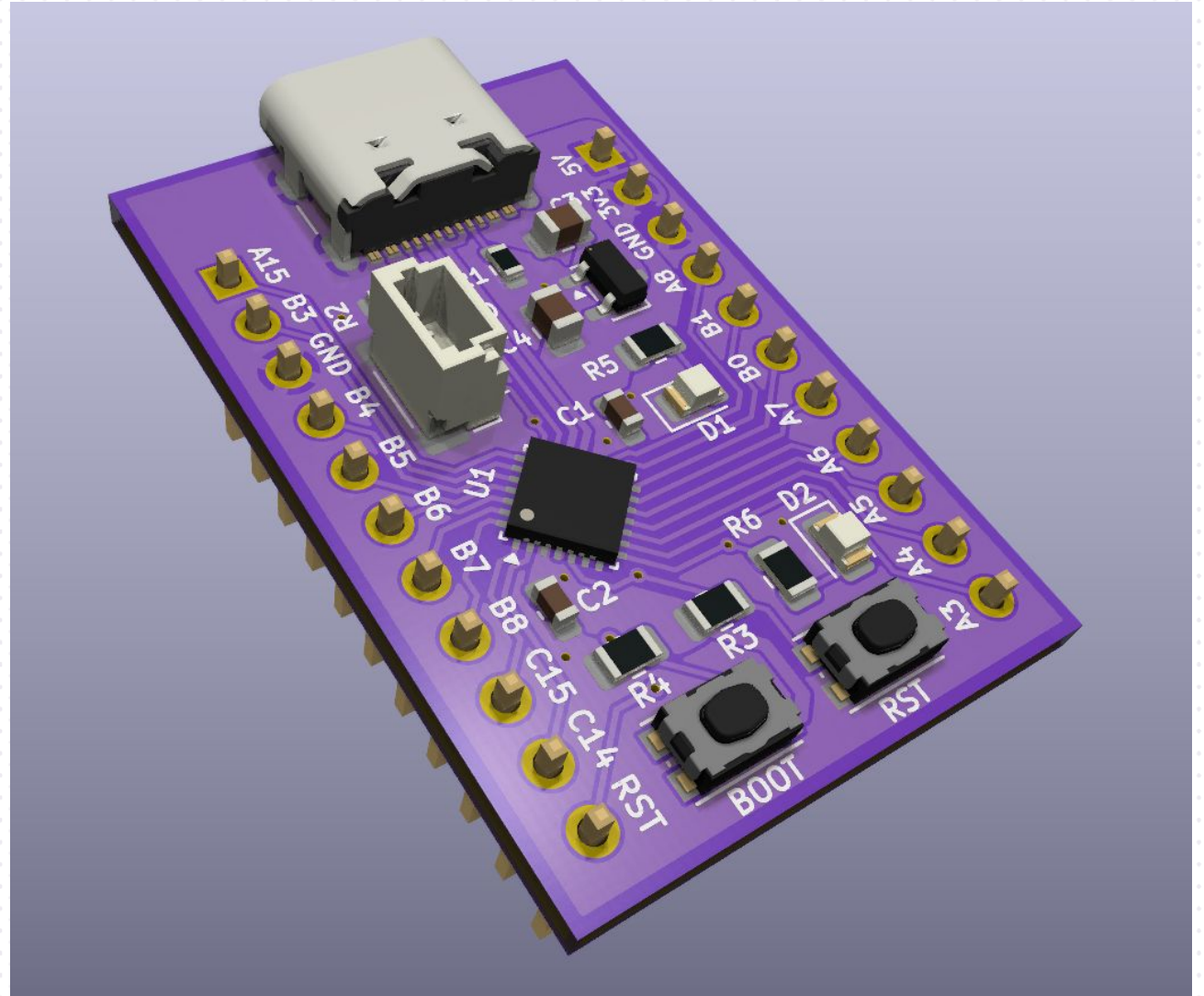
Workshop Goals

- Why and when to use PCBs
- PCB design best practices
- Using PCB design software
- Assembling PCBs

Our Project

STM32 board,
Arduino-compatible

- USB-C
- 17 (kinda 18) GPIO
- Built in LED
- Debug Header

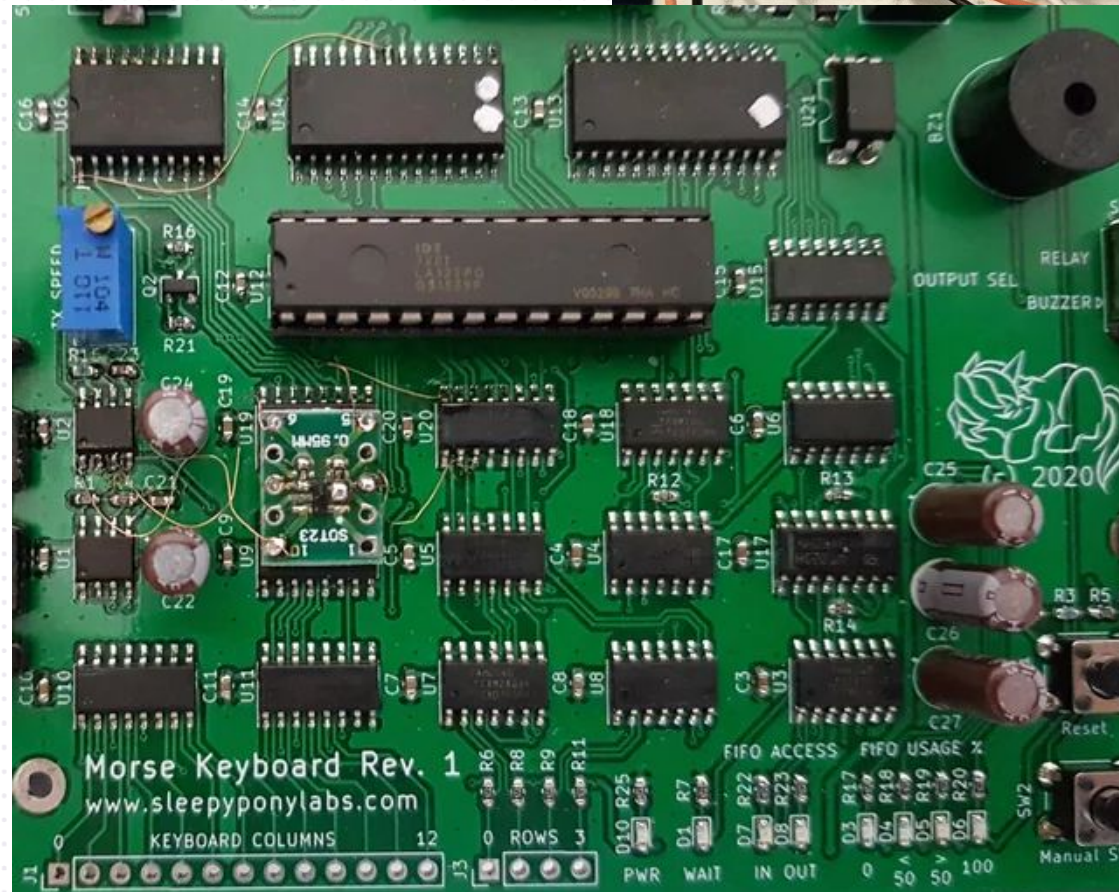
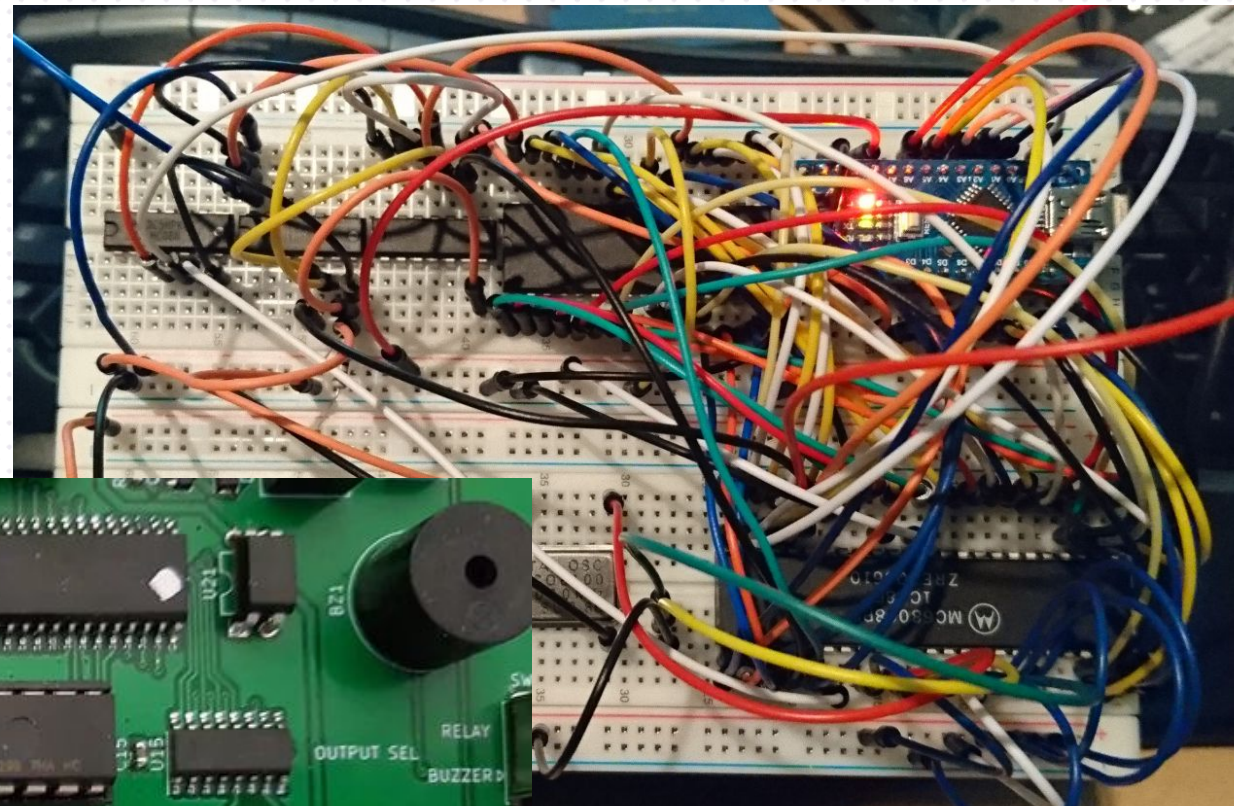


Today's Goals

- Understand why PCBs are important
- Learn the PCB design workflow
- PCB design best practices
- How to install KiCad

Why Use PCBs over Breadboards

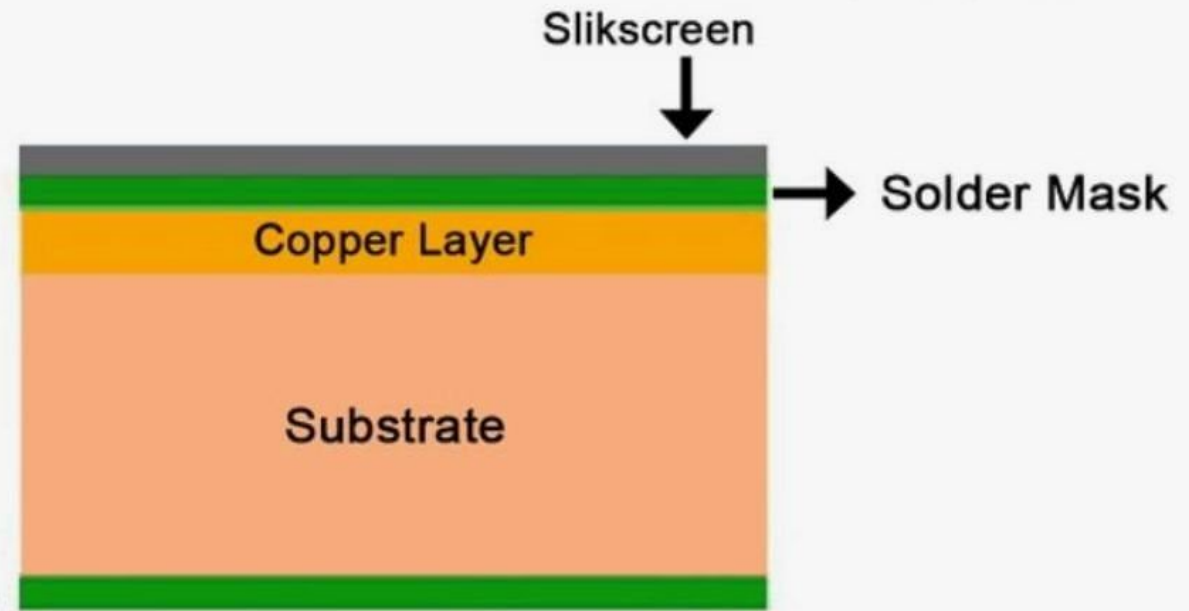
- Takes up less space
- Less likely to fail
- More parts available



Layers of a PCB

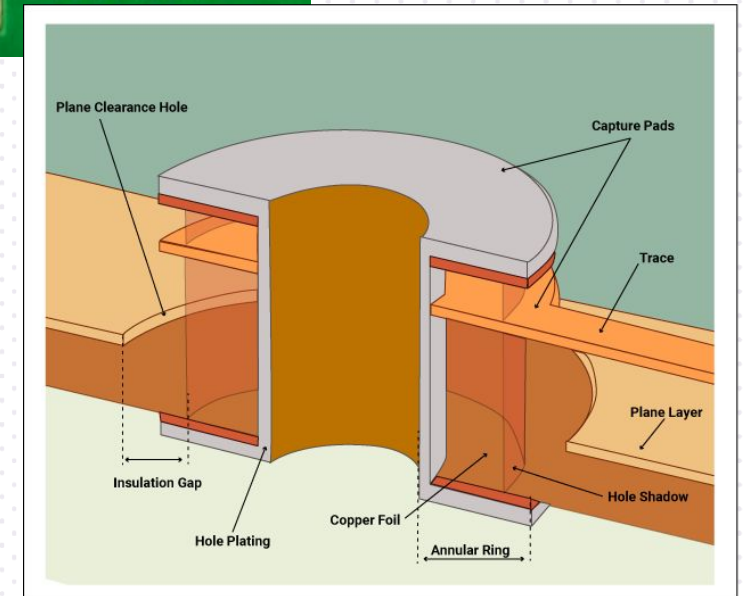
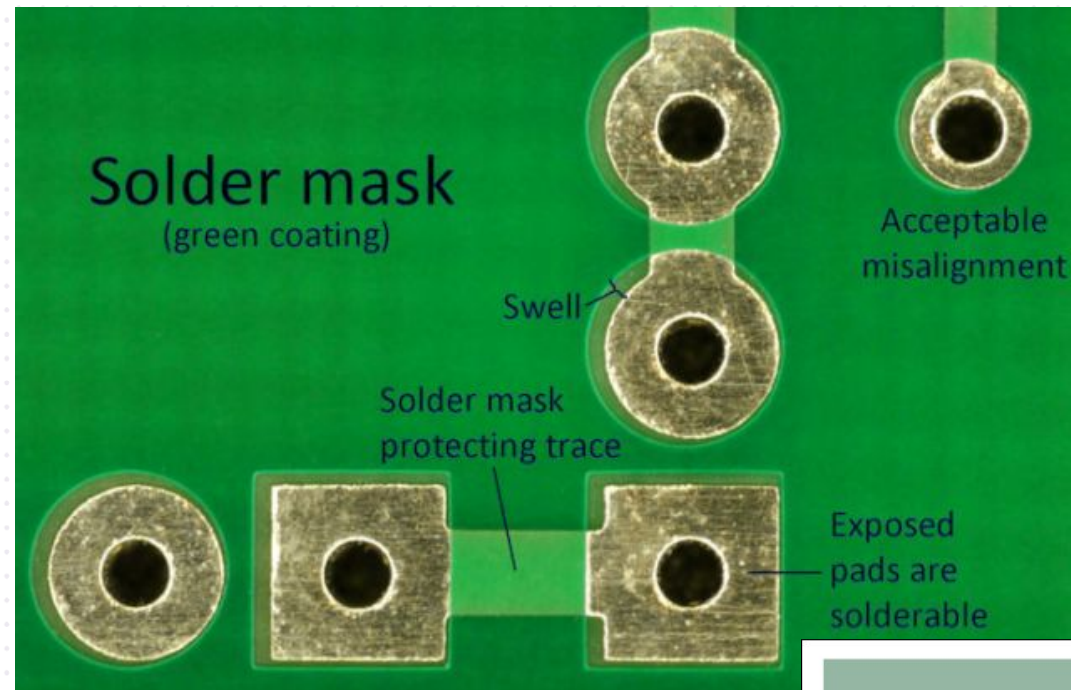
- Copper
- Substrate
- Solder mask
- Silkscreen

PCB Layers



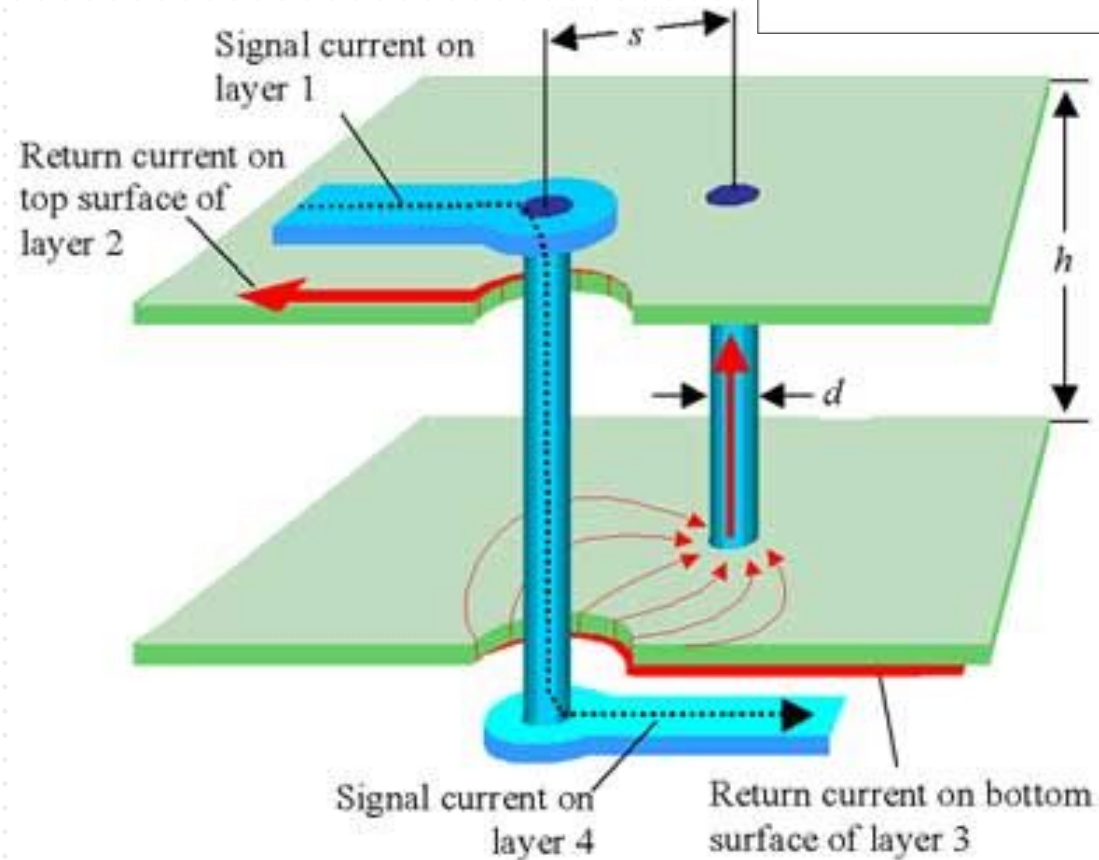
Features on a PCB

- Pad
- Trace
- Via
- Plane
- Footprint



The Electrical Behavior of a PCB

- Return currents (low vs high speed)
- Current loops
- Parasitic capacitance



PCB Design Workflow

Requirements & Schematic

- Figure out what the board does
- Figure out how to do that thing
- Plan out how to connect everything in order to do that thing

Layout & Routing

- Defines the physical size of the board
- Plan the layout so routing is easy
- Route traces to connect everything together

DRC & Fab Outputs

- Verify that your design can be made
- Export the design files for sharing

Order & Manufacturing

- Upload exported design files
- Fill out requested specs and features
- Place order

Assembly & Testing

- Place and solder components
- Test for shorts and connections
- Upload any code, verify functionality

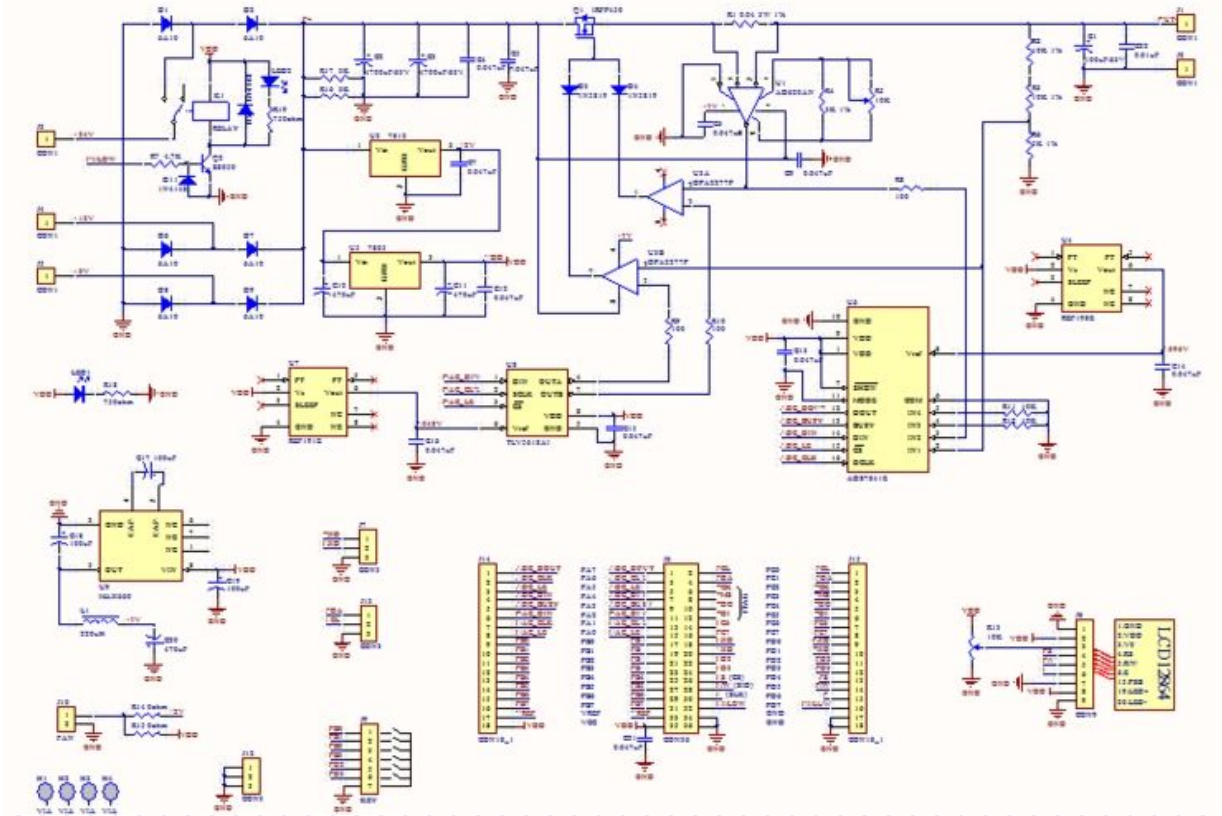
Schematics

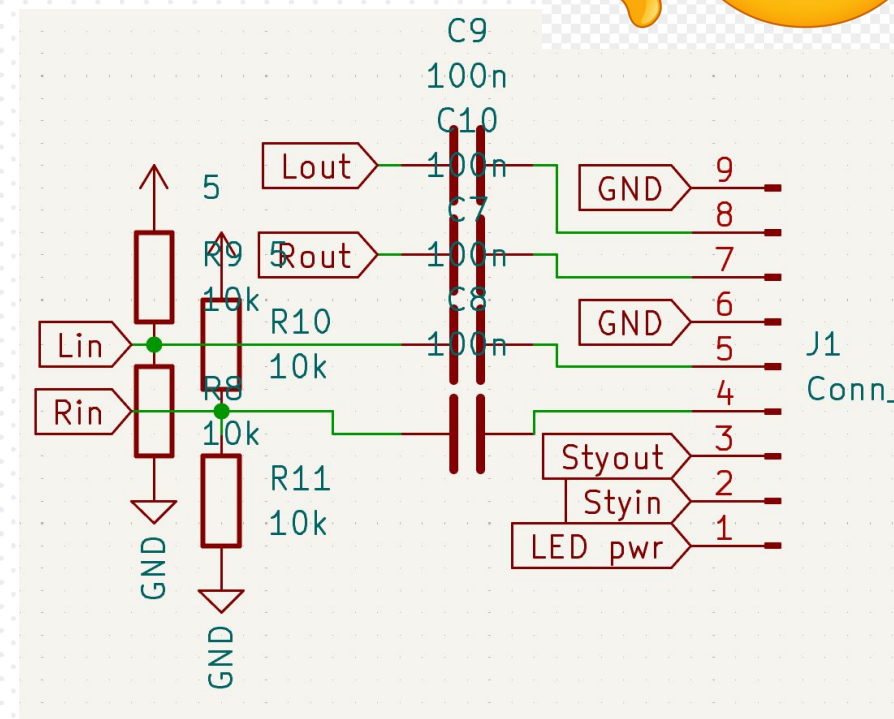
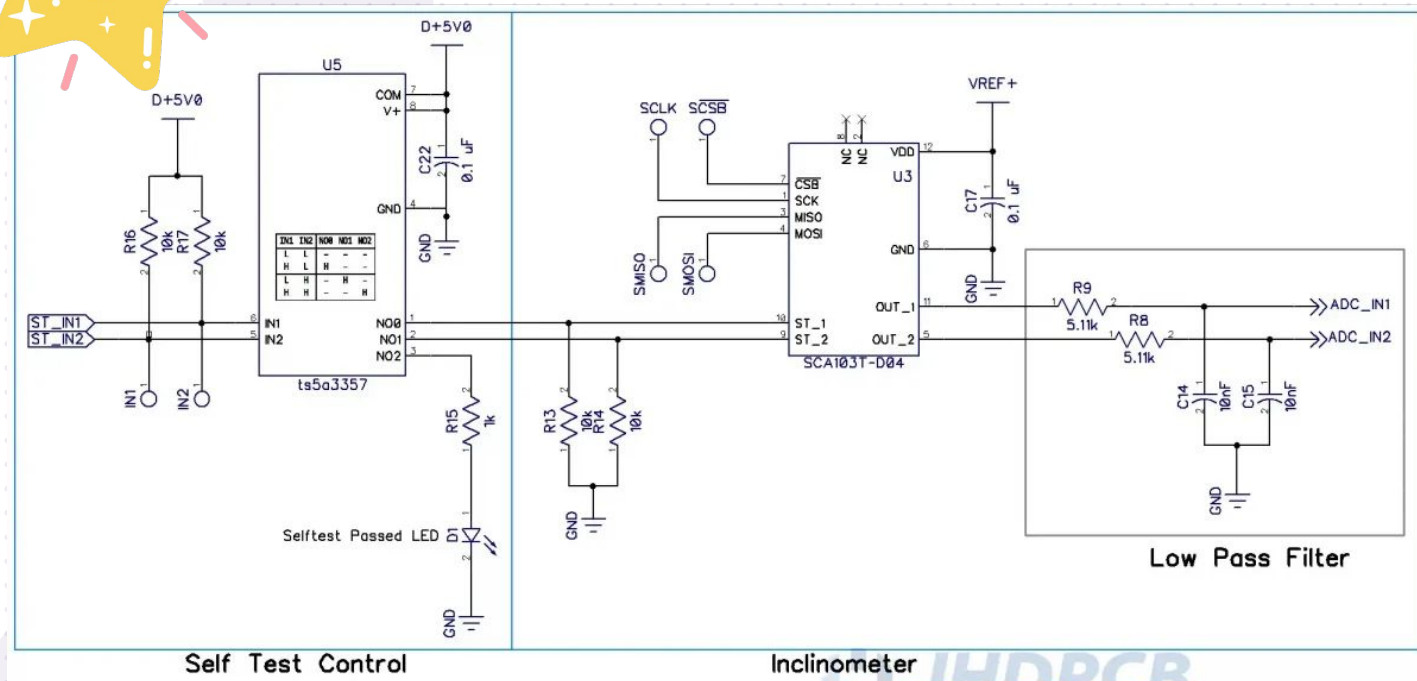
Left to right flow

Power is up, ground is down

Group by function

Documentation!





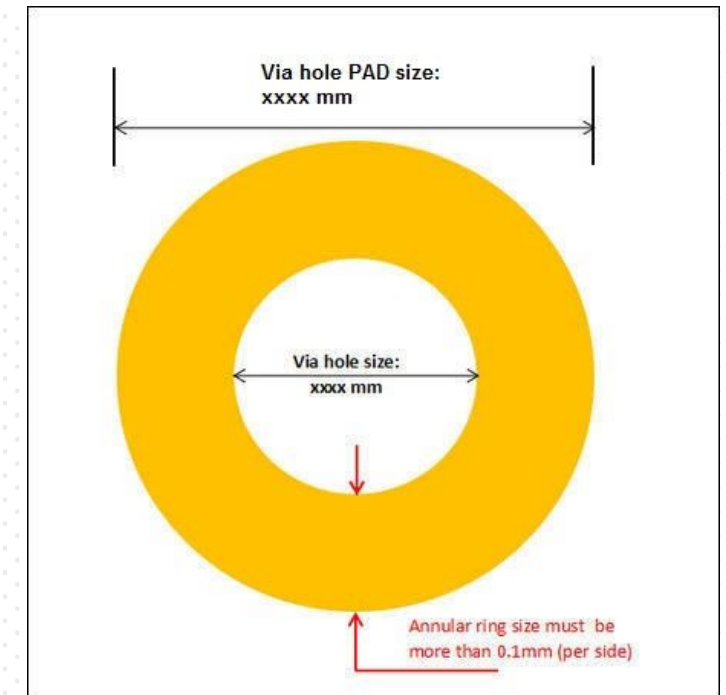
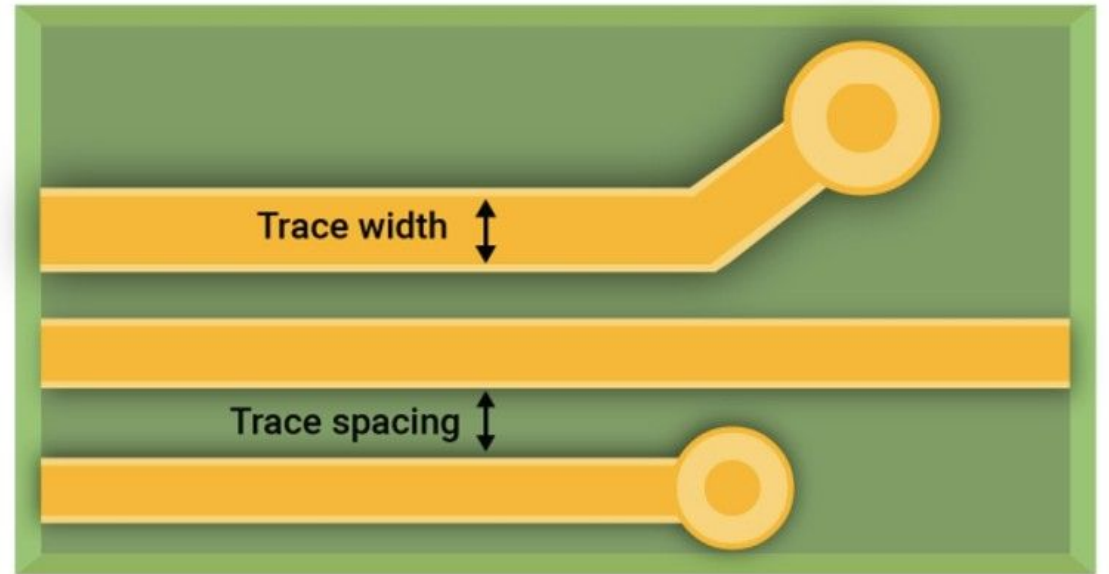
JHDPCB

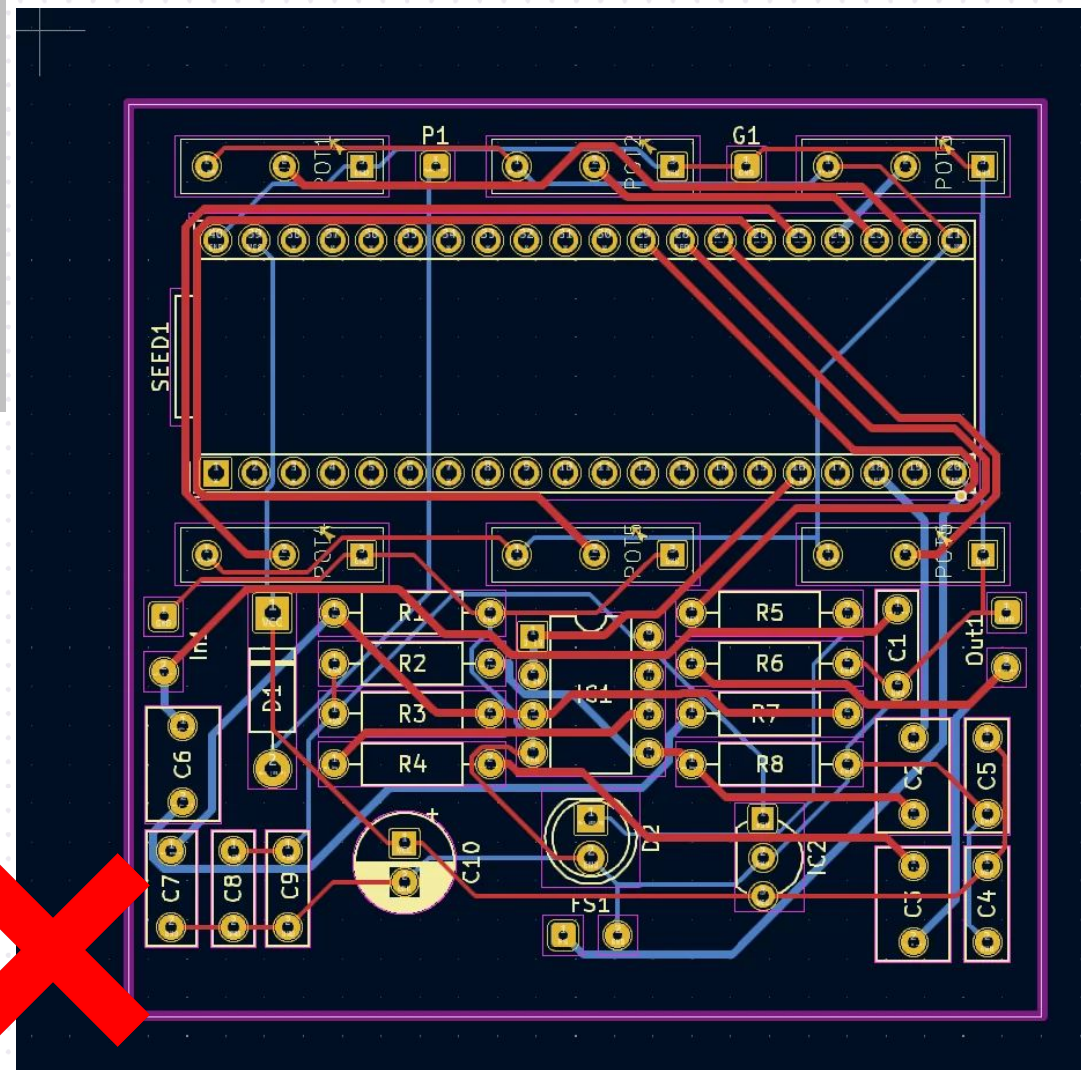
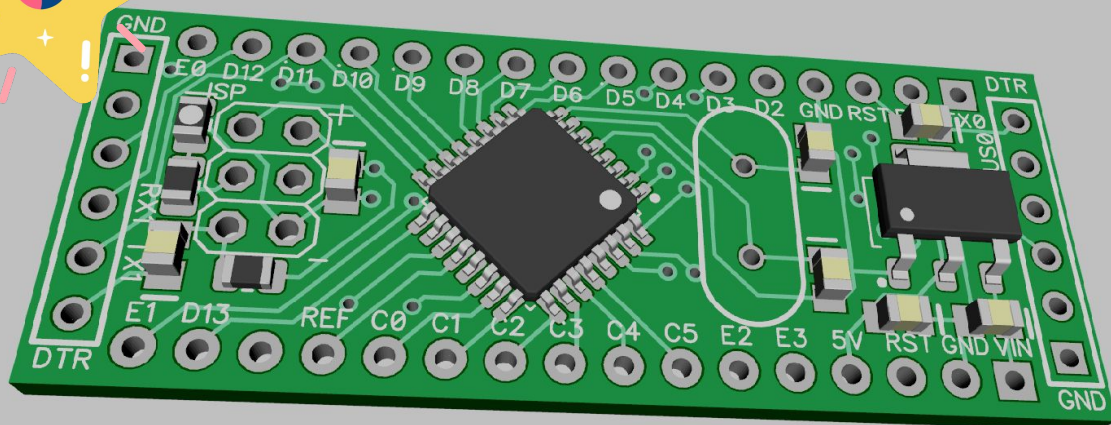
Design Rules

Always set these up!

- Minimum trace width/spacing
- Minimum via size
- Clearance requirements
- Etc

Minimums are just minimums!
Typically you want to use greater
than the minimum



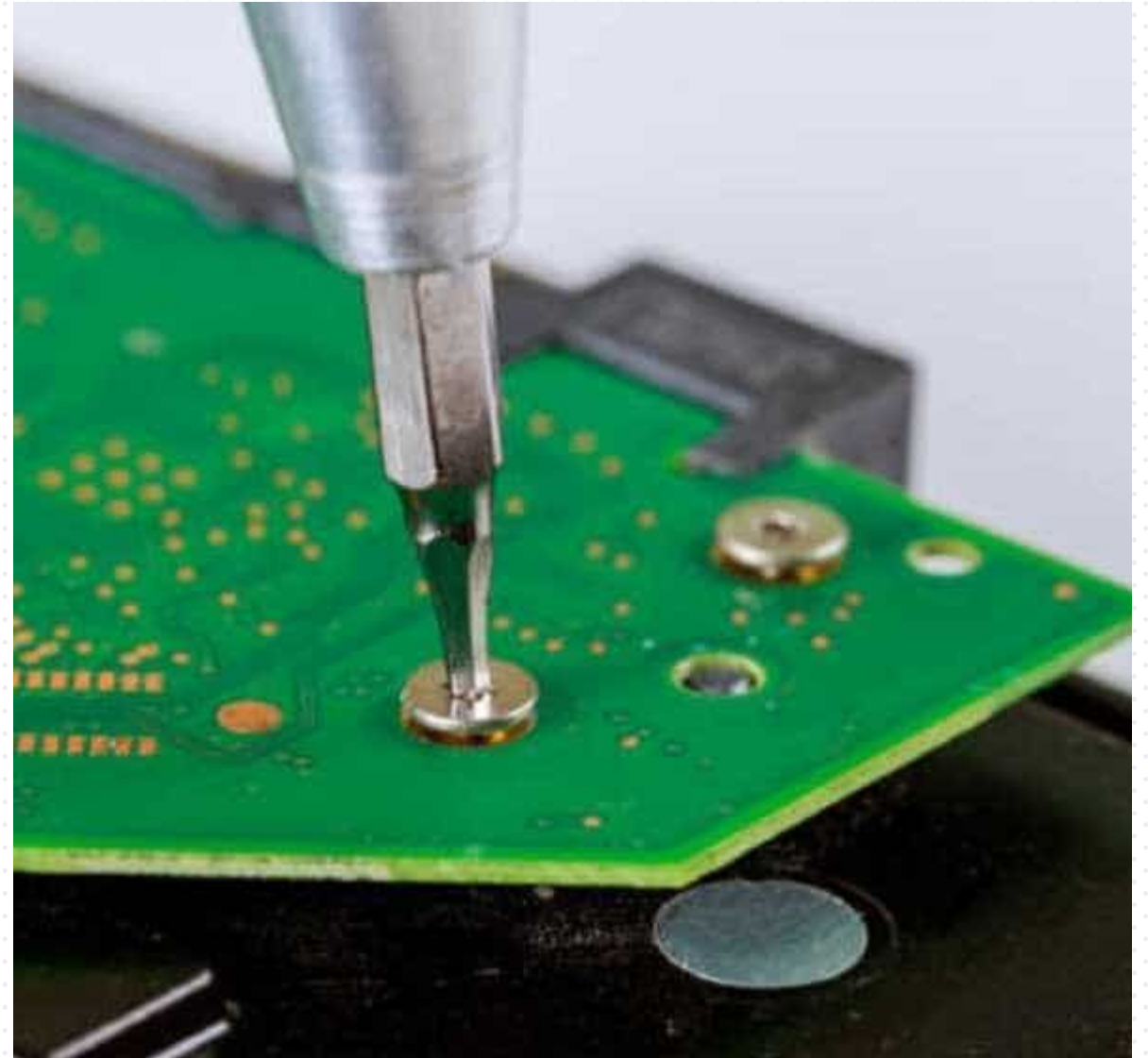


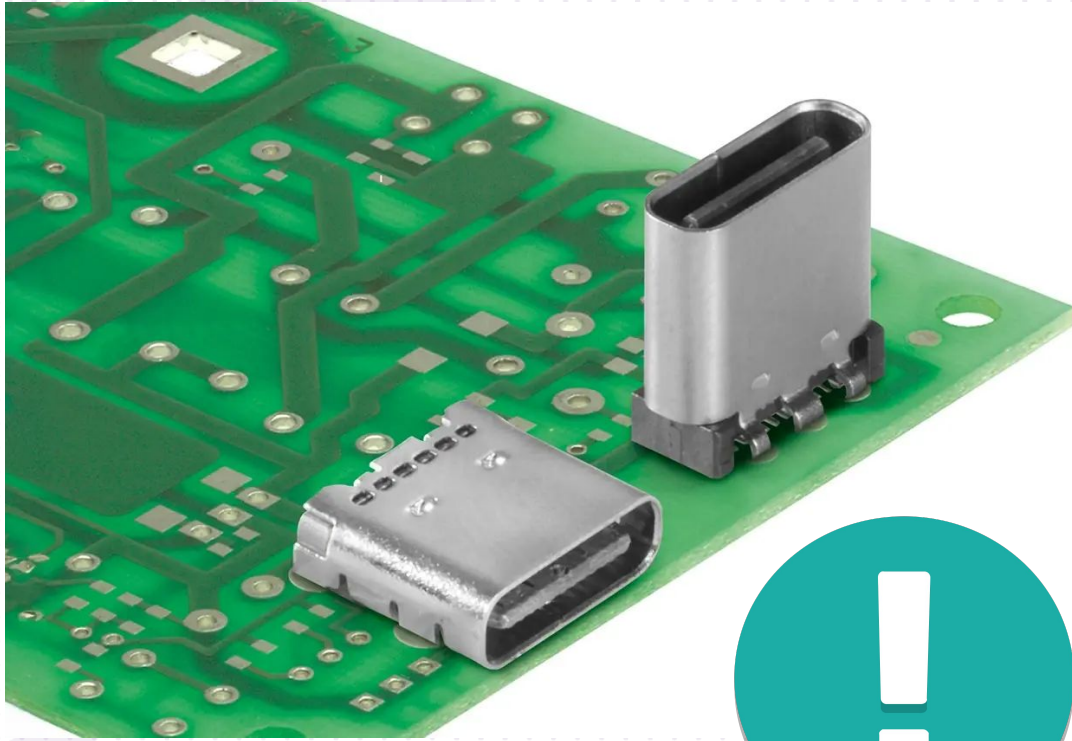
Physical Aspects

Outline/edge cuts

Connector placement

Mounting holes



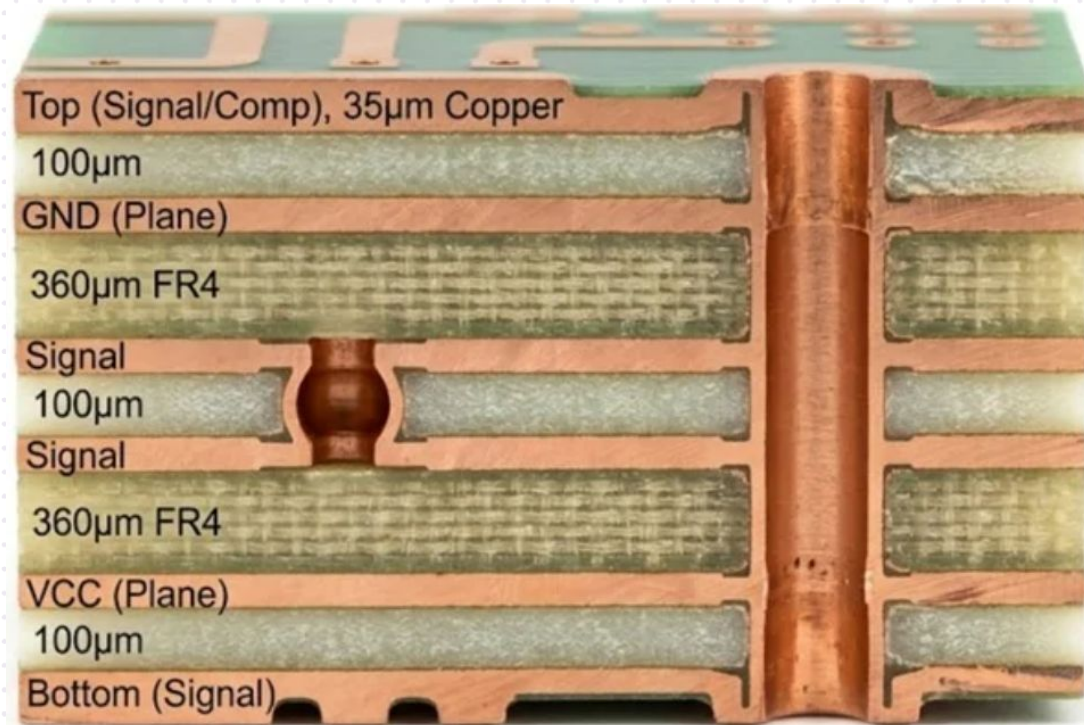


Stack up

Number of layers

Purpose of each layer

Common stackups:



2 Layer

Sig+PWR
GND

Sig+PWR
Sig+GND

4 Layer

Sig
GND
GND
Sig

Sig
GND
PWR
Sig (slow)

6 Layer

Sig
GND
PWR
Sig
GND
Sig

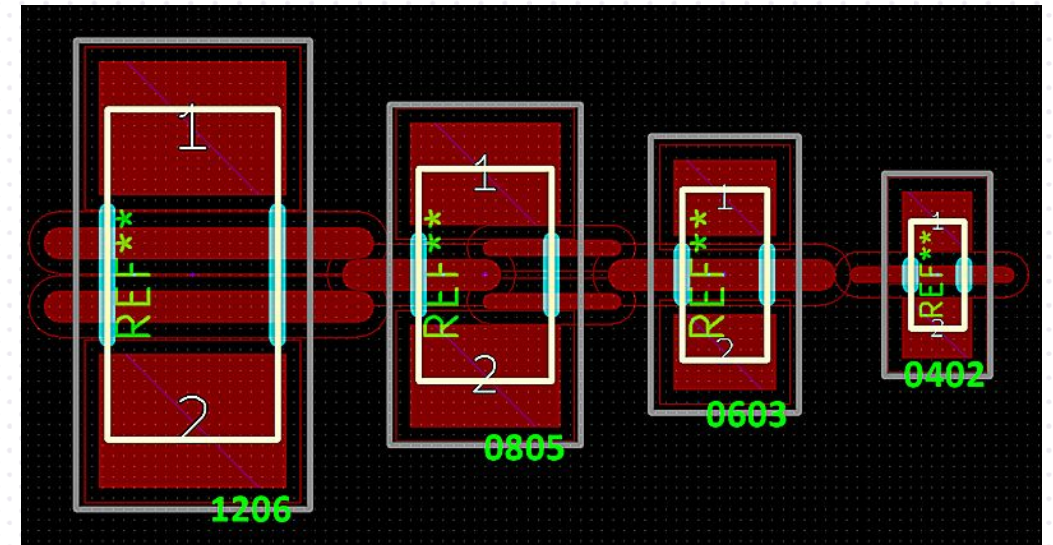
Sig
Gnd
Sig
Sig
Pwr
Sig

Footprint/Component Selection

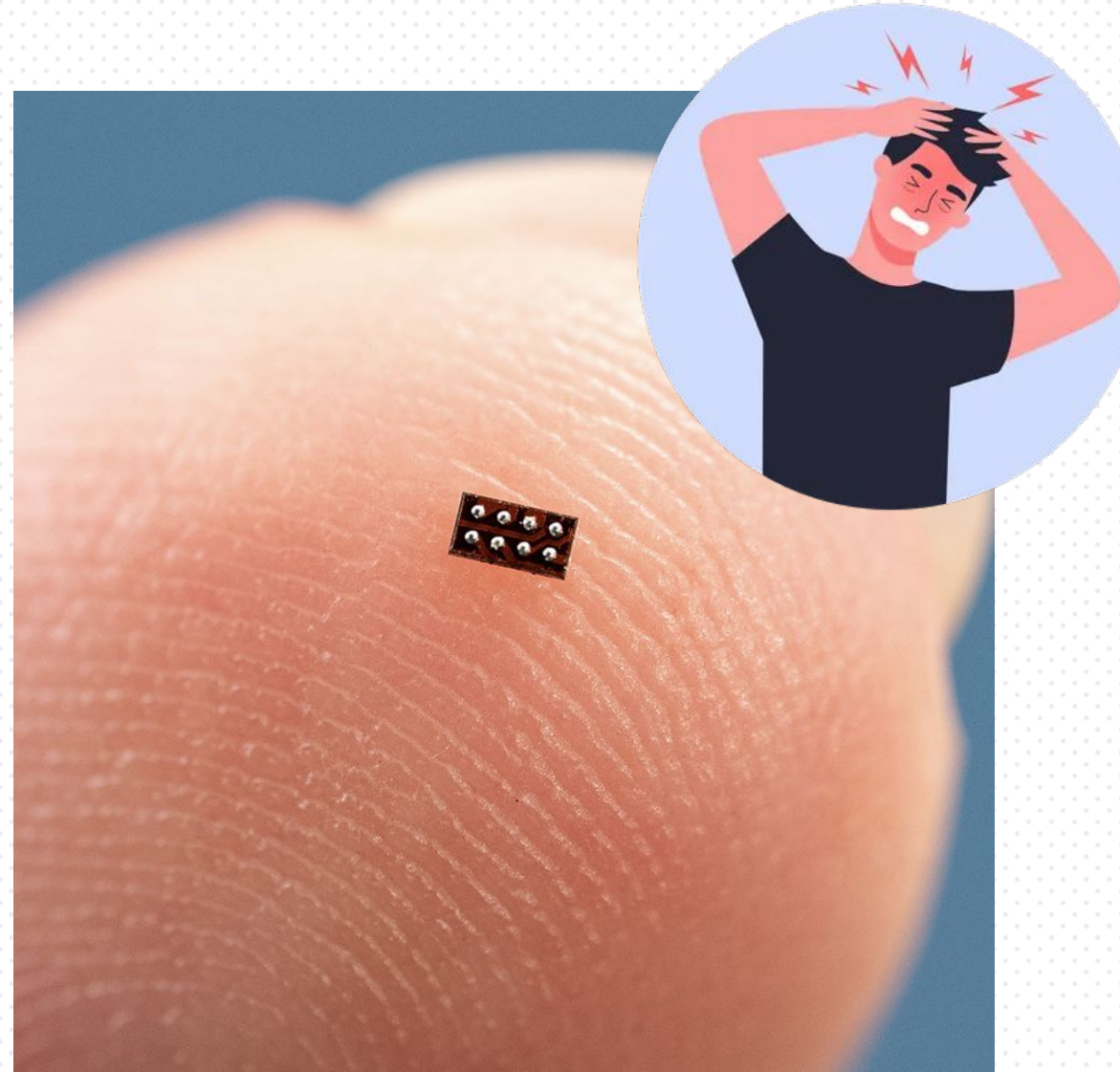
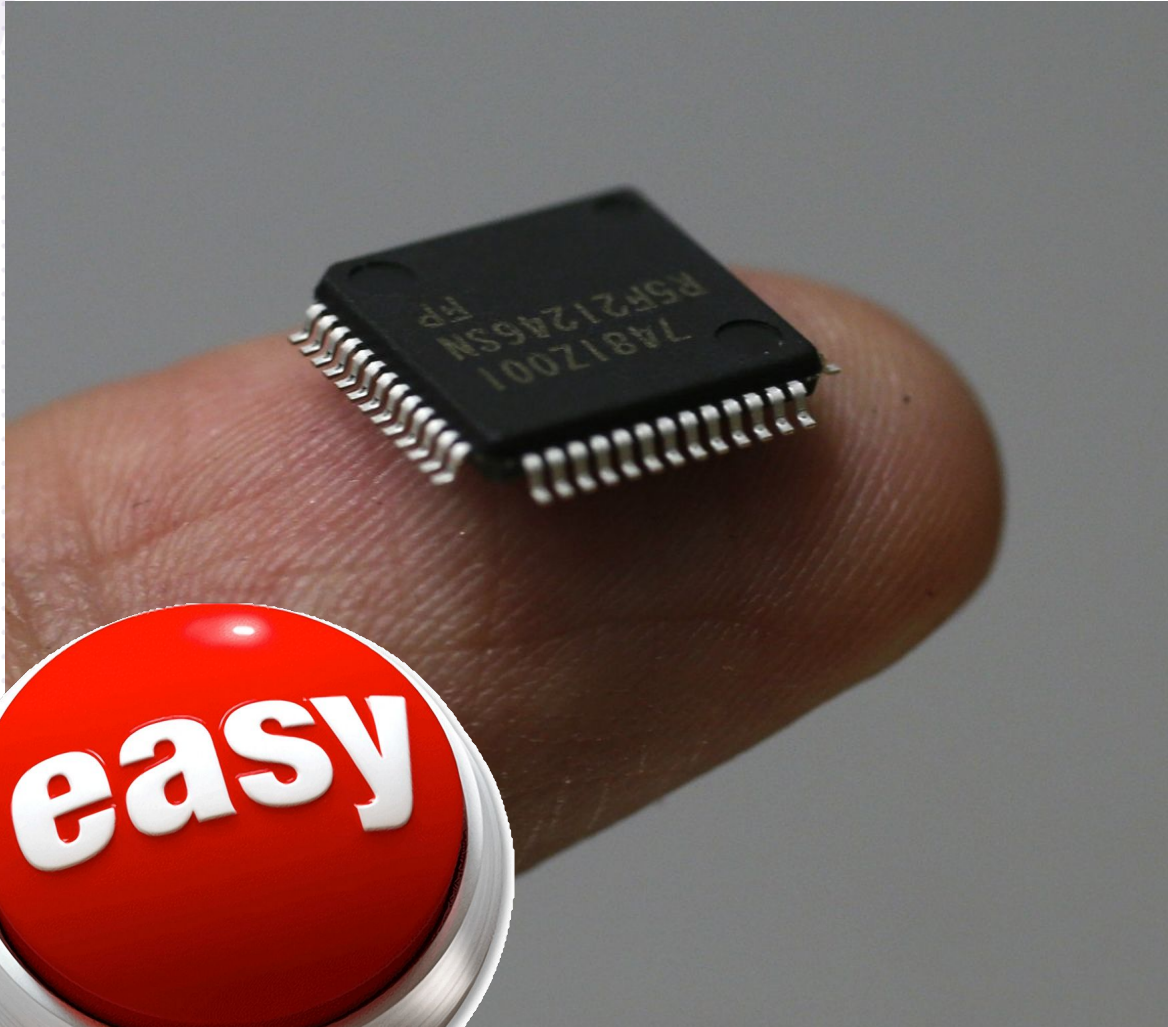
Choose appropriate footprints

Check dimensions from against the data sheet

Check pin orders on multi-pin parts



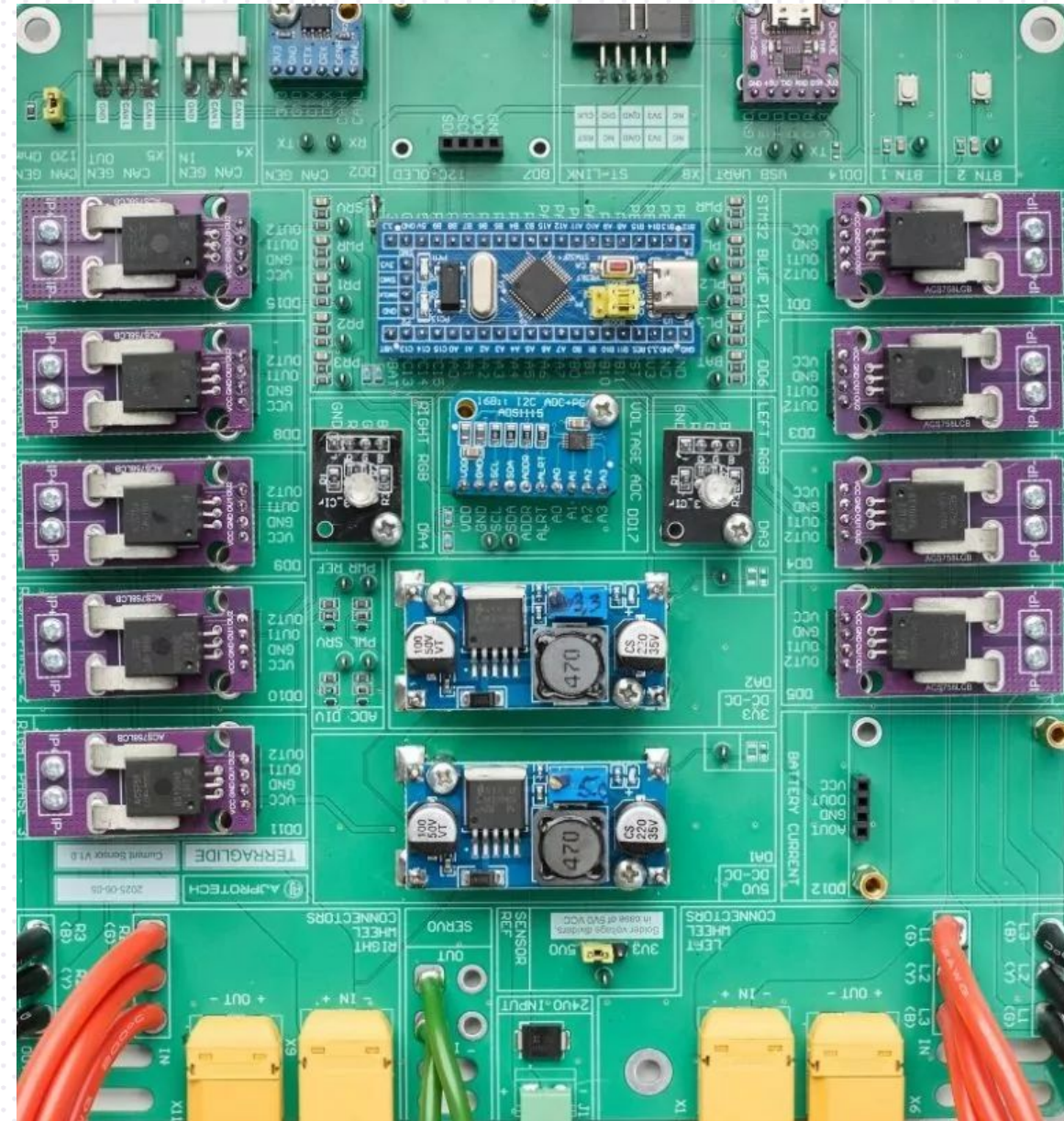
Note: The number refers to the dimensions of the component (ex. 0603 => 60x30 thousandths of an inch)

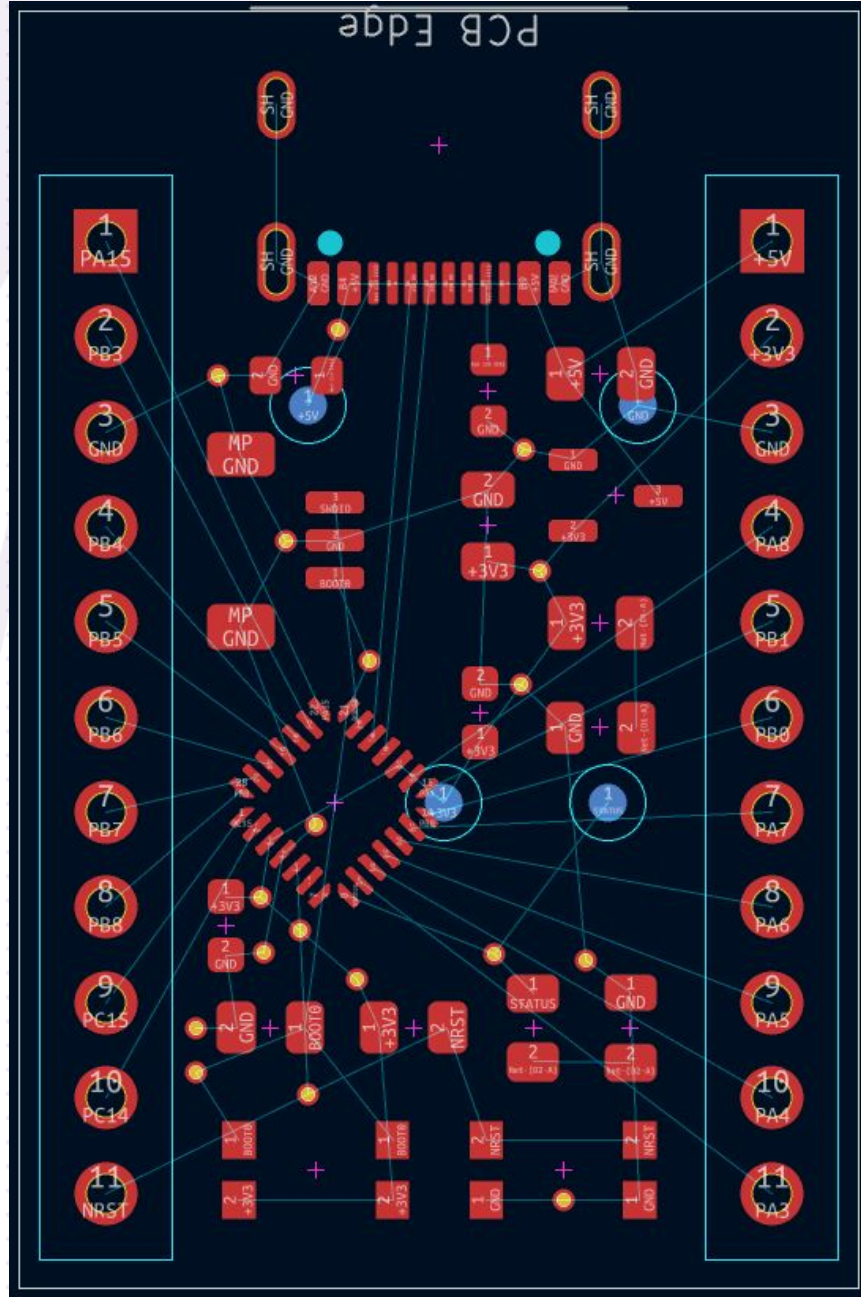
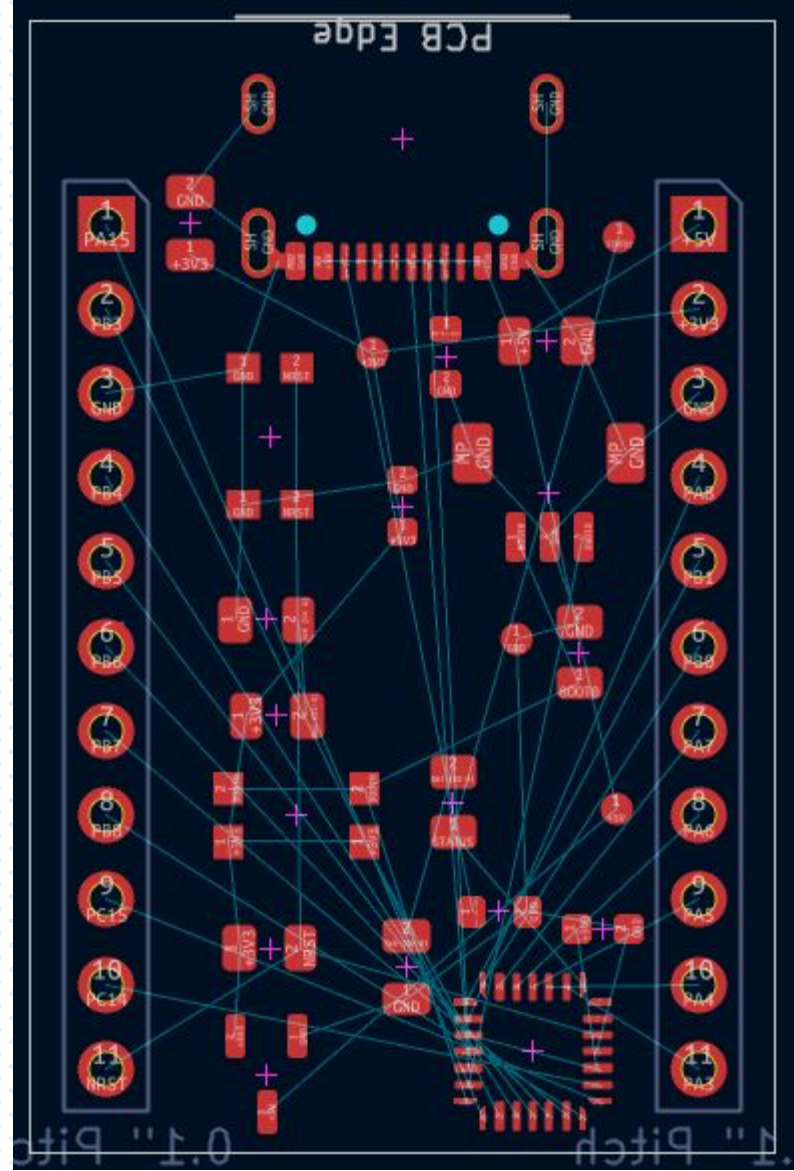


Layout

- Placement by functional groups
- Decoupling caps
- Minimize ratline crossing

[Falstad Simulation](#) of decoupling capacitor





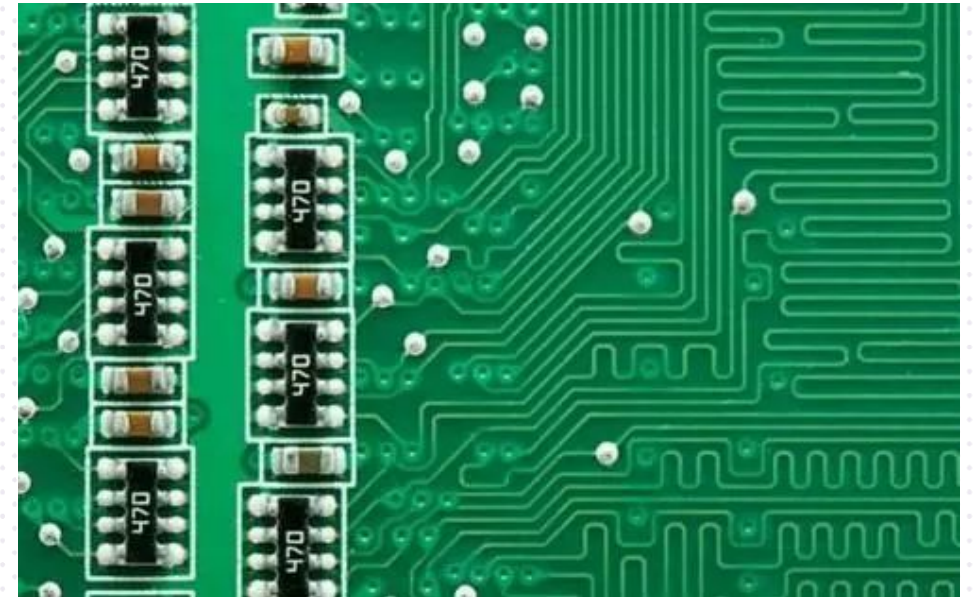
Power Distribution

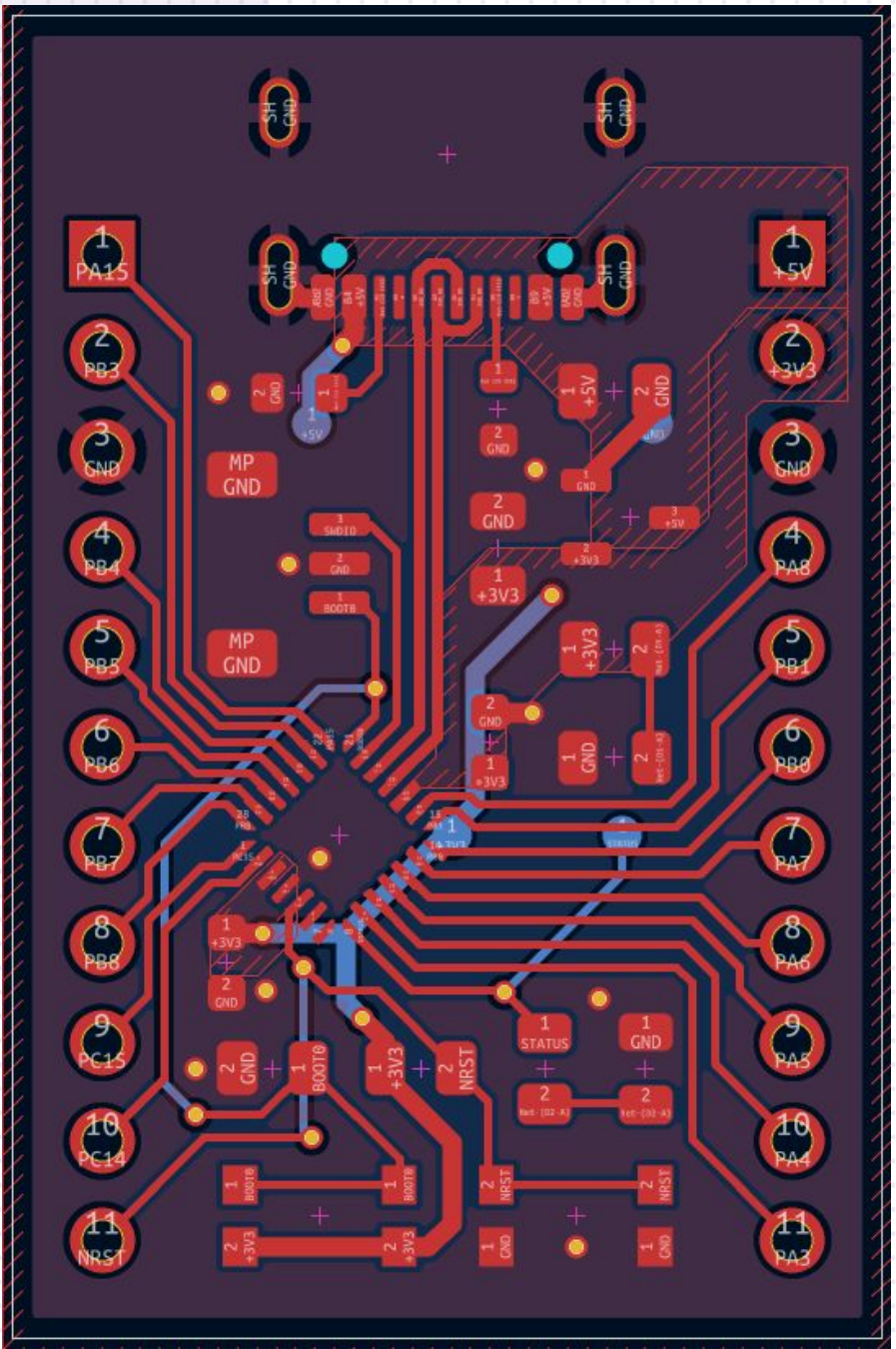
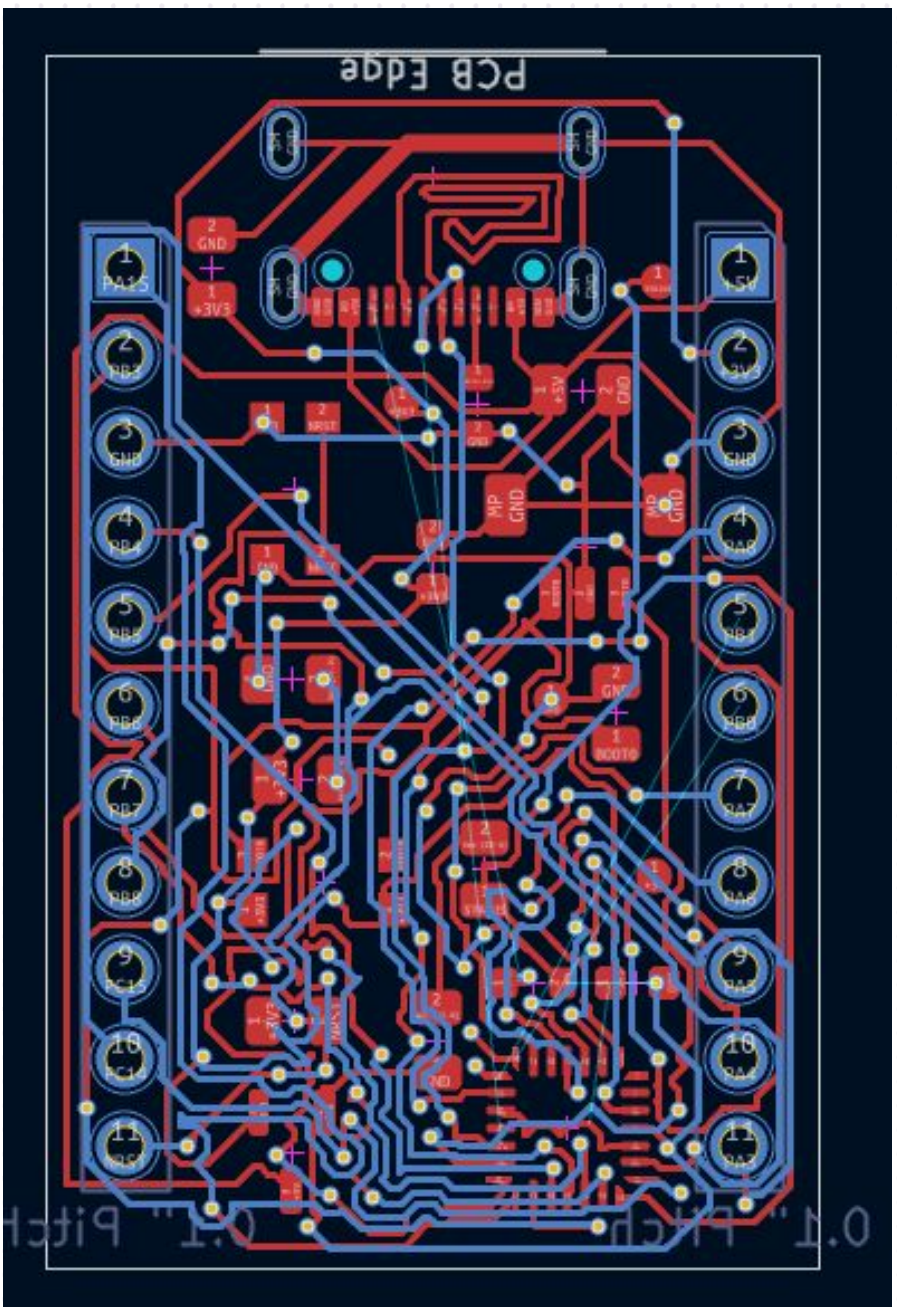
- Solid ground plane whenever possible
- Solid power plane when you can
- Star instead of daisy chain power distribution
- (again) Decoupling capacitors!



Signal Integrity and Reliability

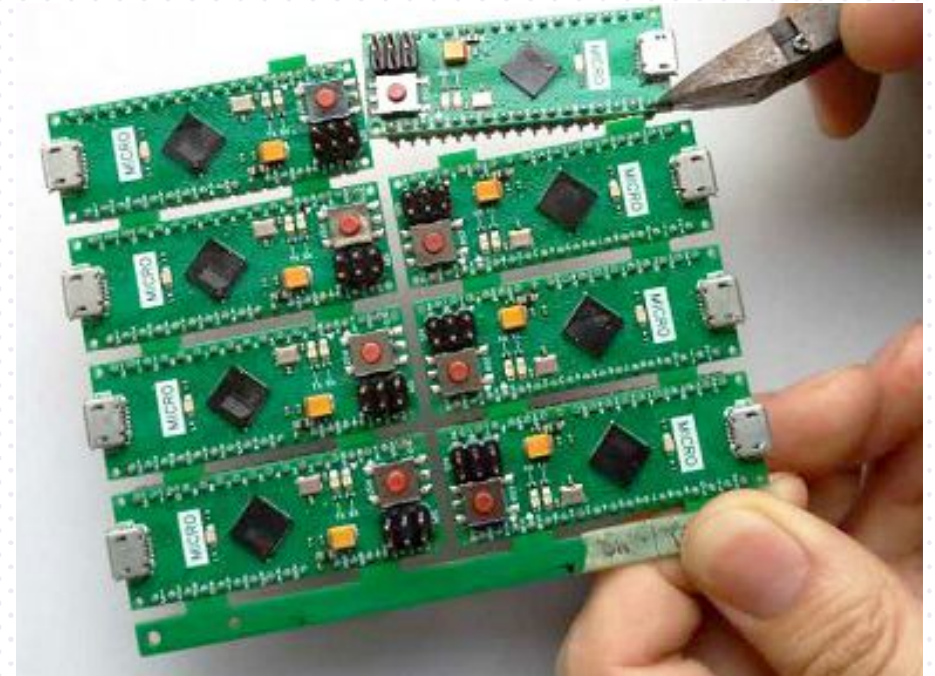
- Decoupling caps close to all ICs (this is important)
- Reference plane for high speed signals
- Place and route most sensitive signals first
- Impedance/length matching





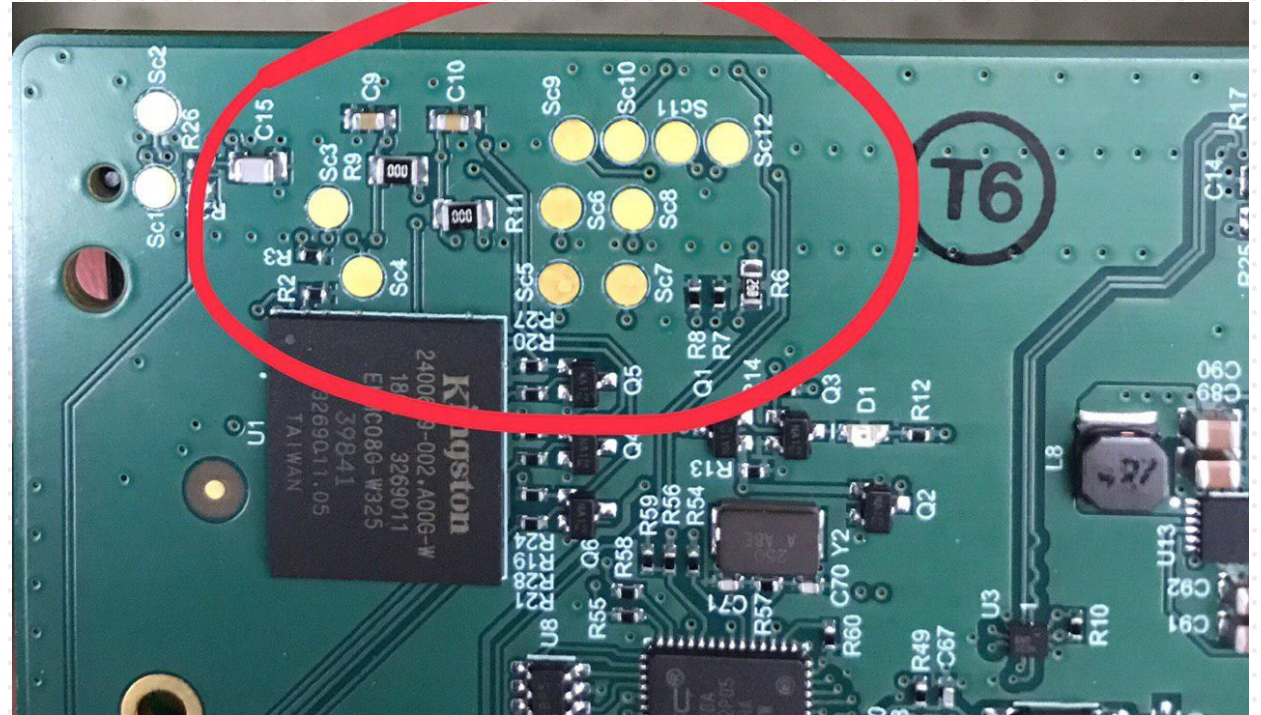
Manufacturability

- No vias in pads!
- Have some space between components
- Aim for one-sided designs



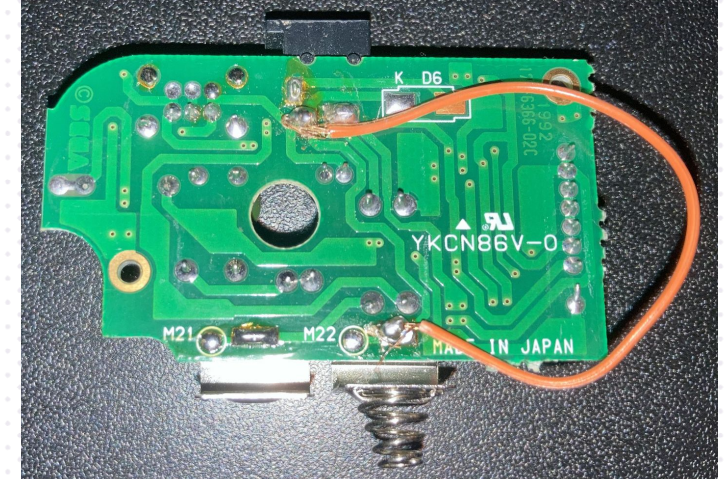
Debuggable Designs

- Lavish use of silkscreen (it's free!)
- Test points (it's free!)
- Power/Status LEDs (it might as well be free!)



Flexible Designs

- Not sure if you need a part? Add it anyway
(Footprints are free!)
- Extra pins? Break them out
(Footprints are free!)
- Thicker signal traces, routed on the outside
(It's easier to bodge thicker traces)



Before next session!

Download and install KiCad 10

(<https://www.kicad.org/download/>)

Session 2 (6 Oct) and Session 3 (8 Oct)
are both working with KiCad

PCB assembly will be during Session 4
(13 Oct)



Registration for session 2

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